MAPS BETWEEN FINITE LOCALLY FREE COMPLEXES ARE REPRESENTED VIA BOUNDED LOCALLY FREE ROOF UNDER THE RESOLUTION PROPERTY

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Allowed standard prerequisites (editorial).

- A quasi-compact quasi-separated scheme with the resolution property has affine diagonal. invoked at proof lines 9928-9929. Enables the representation proposition.
- On a scheme with affine diagonal a morphism of perfect objects is represented by a roof with quasi-coherent middle term. invoked at proof lines 9930-9938. Produces the roof E <- H -> F with H quasi-coherent and H -> E a quasi-isomorphism.
- Because E and F are bounded, the roof maps factor through a truncation of H, allowing H to be replaced by a bounded below complex. used at proof lines 9938-9946. Prepares the input for the strictly perfect construction.
- The strictly perfect construction lemma replaces a bounded below quasi-coherent middle term by a bounded complex of finite locally free modules. invoked at proof lines 9947-9948. Concludes.

Source reference. Original lemma 0F8C: sources/stacks-perfect/pinned-original-source.tex, source lines 9705–9705. Cited at proof lines 9928-9929 for the affine diagonal; the roof construction is inside the counted span.

Source reference. Original proposition 08DB: sources/stacks-perfect/pinned-original-source.tex, source lines 1801–1801. Cited at proof lines 9930-9931 to represent alpha by a quasi-coherent roof; the truncation and conclusion are inside the counted span.

Source reference. Original lemma 0F8E: sources/stacks-perfect/pinned-original-source.tex, source lines 9790–9790. Cited at proof lines 9947-9948 as the final step; the boundedness and truncation arguments are inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Let $\mathcal{E}^\bullet$ and $\mathcal{F}^\bullet$ be finite complexes of finite locally free $\mathcal{O}_X$-modules. Then any $\alpha \in \mathrm{Hom}_{D(\mathcal{O}_X)}(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ can be represented by a diagram*

$$\mathcal{E}^\bullet \leftarrow \mathcal{G}^\bullet \rightarrow \mathcal{F}^\bullet$$

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where $\mathcal{G}^\bullet$ is a bounded complex of finite locally free $\mathcal{O}_X$-modules and where $\mathcal{G}^\bullet \to \mathcal{E}^\bullet$ is a quasi-isomorphism.

Proof. By Lemma 0F8C we see that X has affine diagonal. Hence by Proposition 08DB we can represent α by a diagram

$$\mathcal{E}^\bullet \leftarrow \mathcal{H}^\bullet \to \mathcal{F}^\bullet$$

where $\mathcal{H}^\bullet$ is a complex of quasi-coherent $\mathcal{O}_X$-modules and where $\mathcal{H}^\bullet \to \mathcal{E}^\bullet$ is a quasi-isomorphism. For $n \ll 0$ the maps $\mathcal{H}^\bullet \to \mathcal{E}^\bullet$ and $\mathcal{H}^\bullet \to \mathcal{F}^\bullet$ factor through the quasi-isomorphism $\mathcal{H}^\bullet \to \tau_{\geq n}\mathcal{H}^\bullet$ simply because $\mathcal{E}^\bullet$ and $\mathcal{F}^\bullet$ are bounded complexes. Thus we may replace $\mathcal{H}^\bullet$ by $\tau_{\geq n}\mathcal{H}^\bullet$ and assume that $\mathcal{H}^\bullet$ is bounded below. Then we may apply Lemma 0F8E to conclude. □